

Problem set #2

MATH 4503/6503: Numerical Methods for Differential Equations

Due date: February 16, 2026

Reading resources: Notes uploaded to D2L; book pages 113-115 and 120-128

Problem #1: $y'(x) = f(x, y(x))$ as a system of equations

Welcome! You have made it to the second problem set. In the last couple of weeks, we have developed some numerical stencils in order to solve $y'(x) = f(x, y(x))$. I mentioned in class that the same stencils could be used even if $y'(x) = f(x, y(x))$ represented a system of equations. The aim of this problem is to put that claim to good use. Let us dive right in!

We are given that

$$y'(x) = \begin{bmatrix} y_1'(x) \\ y_2'(x) \end{bmatrix} = \begin{bmatrix} y_2(x) \\ -y_1(x) \end{bmatrix} = \begin{bmatrix} f_1(x, y(x)) \\ f_2(x, y(x)) \end{bmatrix} = f(x, y(x)).$$

Start by writing the trapezoidal stencil for this system of equations. At this point, keep $f_1(x, y(x))$ and $f_2(x, y(x))$ completely general; that is, do not substitute $y_2(x)$ and $-y_1(x)$ for them, respectively. Your answer should simply show the relationship between $Y_{1,i+1}(\approx y_1(x_{i+1}))$, $Y_{2,i+1}(\approx y_2(x_{i+1}))$ and $Y_{1,i}(\approx y_1(x_i))$, $Y_{2,i}(\approx y_2(x_i))$. (*Hint:* You might want to use vector or matrix data structures in Maple here.)

- (a) We know that the trapezoidal stencil is implicit; however, the form we have chosen for $f(x, y(x))$ is simple enough to allow an explicit stencil to be written. Write that stencil in this part.
- (b) Use the stencil in part (a) to solve the given system with initial conditions $y_1(0) = 1, y_2(0) = 0$ for $x \in [0, 5]$. Work with $N = 100$ steps.
- (c) We performed a discrepancy analysis for the trapezoidal method in class but with $y'(x) = f(x, y(x))$ being a single equation rather than a system. Use `dsolve` to find the analytic solution to the given system with the initial conditions in part (b). Then, perform the full discrepancy analysis for the system to verify that the global error (take it to mean the error at $x = 5$) for the trapezoidal method is $O(h^2)$.

Hint: Error at $x = x_i$ may be computed as $\sqrt{(Y_{1,i} - y_1(x_i))^2 + (Y_{2,i} - y_2(x_i))^2}$.

Problem #2: a centered method, hmm?

We developed forward Euler, backward Euler, and trapezoidal methods for $y'(x) = f(x, y(x))$ in class; the forward and backward methods employed the one-sided approximations $D_+y(x)$ and $D_-y(x)$, respectively, to give us the corresponding stencils, and the trapezoidal was just their average. Could we use the centered approximation

$$D_0y(x) = \frac{y(x+h) - y(x-h)}{2h}$$

to build a stencil as well? Yes! So, let us explore this possibility in this problem.

The method we get from $D_0y(x)$ is called the midpoint method (or "leapfrog" method, a name I leave you with pondering the reason for), and the corresponding stencil is obtained as follows:

$$\begin{aligned} y'(x) &= f(x, y(x)) \\ \implies \frac{y(x+h) - y(x-h)}{2h} &= f(x, y(x)) + O(h^{p-1}) \\ \implies y(x+h) - y(x-h) &= 2hf(x, y(x)) + O(h^p) \quad (*) \\ \implies y_{i+1} - y_{i-1} &= 2hf(x_i, y_i) \\ \implies y_{i+1} &= y_{i-1} + 2hf(x_i, y_i), \quad (**) \end{aligned}$$

where $i = 1, \dots, N$, or the total number of steps. Note that in the Euler and trapezoidal methods, we need information at the i th point alone to determine y_{i+1} ; in the midpoint method, however, we need information at the i th and $(i-1)$ th points (see the step marked $(**)$ above). It is thus a multistep method—in contrast to one-step methods, such as the Euler and trapezoidal.

- Consider the step marked $(*)$ above. By employing Taylor series expansions, determine the value of p .
- What does this mean for the expected global error in numerical algorithms using this stencil? (No calculations are necessary; just state the answer in a text block or using a comment in your worksheet.)
- Confirm the expectation of part (b) by using this stencil to numerically solve the initial value problem

$$y'(x) = -y(x), y(0) = 5$$

and then plotting the discrepancy between the numeric and exact values of $y(5)$ against h on a log-log plot.

Hint: We want to be going from $i = 0$ (which represents initial data) through $i = N$ (the total number of steps), but $(**)$ starts from $i = 1$, or $y_2 = y_0 + 2hf(x_1, y_1)$, and we do not have $y_1 \approx y(h)$. You will be required to compute it using one of the one-step methods (I recommend trapezoidal); the rest will be handled by $(**)$.

Problem #3: Taylor series methods

Since we have had some good experience with procedures in Maple, I will not write this problem as a very structured problem, such as problems 1 and 2.

We have discussed in class that the Euler methods can be derived using the Taylor series expansion of $y(x_i + h)$ about $x = x_i$. However, the one-step error in them is $O(h^2)$ as we drop terms with h^2 or higher powers of h . A Taylor series method of higher accuracy can be derived by keeping more terms in the Taylor series. For example, If we keep the first $n + 1$ terms of the Taylor series expansion,

$$y(x_i + h) \approx y(x_i) + hy'(x_i) + \frac{1}{2}h^2y''(x_i) + \dots + \frac{1}{n!}h^ny^{(n)}(x_i),$$

we obtain a method with one-step error $O(h^{n+1})$. The only issue is that we are given just

$$y'(x) = f(x, y(x)),$$

and we must compute the higher derivatives $(y''(x), y'''(x), \dots, y^{(n)}(x))$ by repeated differentiation of this (known) function.

Write a Maple procedure that solves for $y(1)$ in the initial value problem

$$y'(x) = f(y(x)), y(0) = 1$$

using a numerical stencil based on the n th order Taylor series expansion above. The procedure's arguments should include an arbitrary function f , an integer n representing the accuracy of the Taylor series expansion, and another integer N representing the number of steps between $x = 0$ and $x = 1$. Pick a test problem (read, "choose an easy $f(y(x))$ "), and compare your results with the output of `dsolve/numeric`.

Problem #4: Adams-Moulton method (*mandatory for graduate students & bonus for undergraduates*)

Hey, grads (and bonus-seeking undergrads)! We finally have something extra for you to work on. The n -step Adams-Moulton method (another multistep method) to solve $y'(x) = f(x, y(x))$ is defined by

$$y(x + h) = y(x) + h \sum_{j=-1}^{n-1} \beta_j f(x - jh, y(x - jh)) + O(h^{n+2}).$$

Write a Maple procedure with a single argument n that calculates and returns the β_j coefficients. (Do not get too crazy with n ; the amount of time needed for calculation increases very quickly with the number of steps.)